Unit-V

5. (a) What is DOF's? 2

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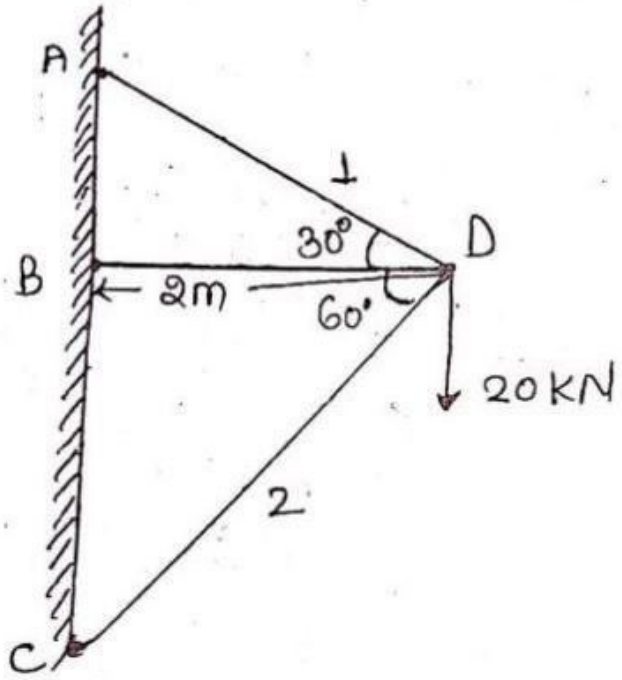
(b) Solve the three bar truss problem in given fig. by minimum potential energy principal. 14



(c) Write application of Reyleign Ritz method & Galerkin method. 14

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B. E. (Eighth Semester) Examination, April-May, 2016

(Old Scheme)

(Civil Engg. Branch)

ANALYSIS of FRAMED STRUCTURES

(Elective-III)

Time Allowed : Three hours

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Maximum Marks : 80

Minimum Pass Marks : 28

*Note : Part (a) is compulsory. Attempt any **one** part of (b) & (c) of each units.*

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Unit-I

1. (a) Define Rotation Factor (RF). 2

(b) In a continuous beam shown in fig.1(b) The support

"B" sink by 10 mm determine the moment by Kani's method and draw BMD & SFD. 14

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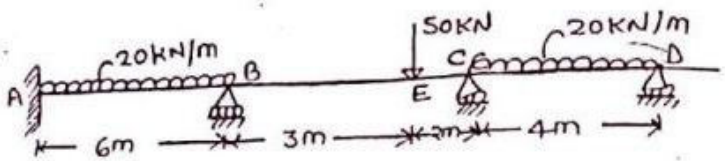

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Fig.-1(b)

Take $I = 1{\cdot}2 \times 10^{-4}\ mm^4$ & $E = 2\times10^5\ N/mm^2$.

(c) Analyse the portal frame & draw the BMD. 14

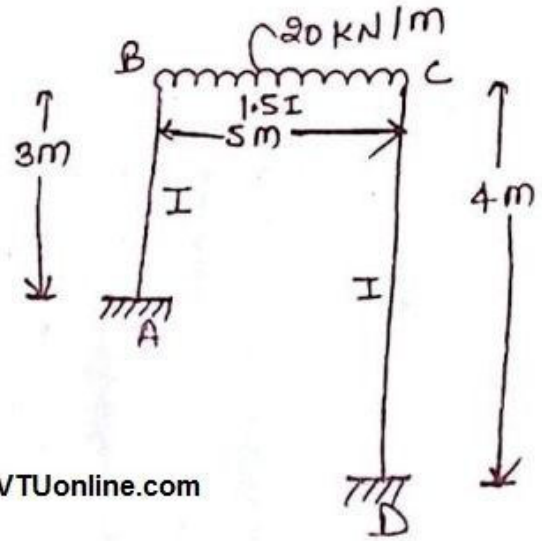

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Fig.-1(c)

Unit-II

2. (a) Write an Assumptions of cantilever & Portal method. 2

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(b) Analyse the building frame, subjected to horizontal force shown in fig.-2(b) using portal method. 14

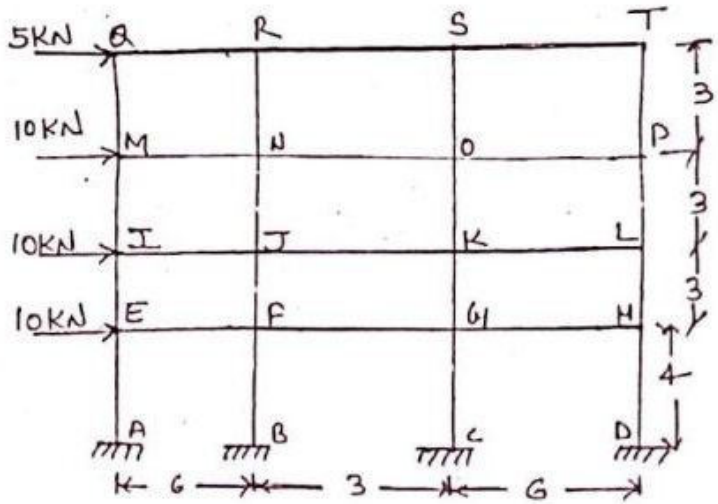

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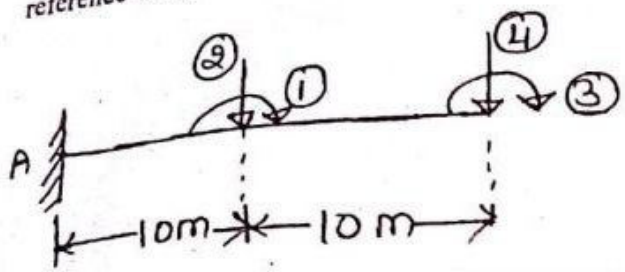
Fig.-2(b)

(c) Analyse the building frame subjected to horizontal force shown in fig.-2(b) using cantilever method. 14

Unit-III

3. (a) Define flexibility matrix. 2

(b) Develop the flexibility matrix for beam AB with reference to the co-ordinates shown in fig. 3(b). 14



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EI = constant

Fig.-3(b)

(c) Analyse the portal frame ABCD shown in fig.-3(c) by flexibility method. 14

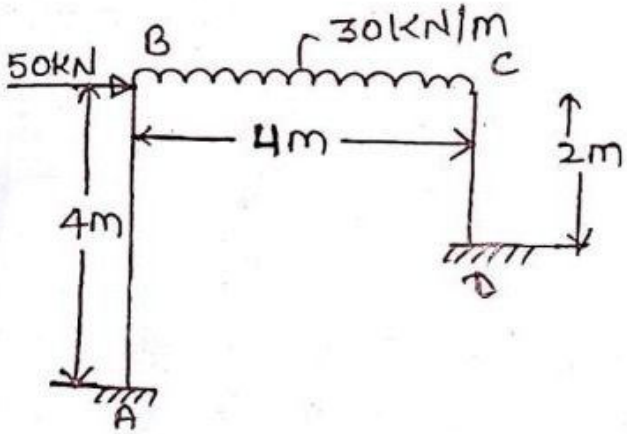

Fig.-3(c)

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Unit-IV

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4. (a) What do you understand by stiffness matrix? 2

(b) Analyse the continuous beam shown in fig.-4(b). If the support B sinks by 10 mm. Use displacement method.

Take EI = 6000 kN-m^2 14

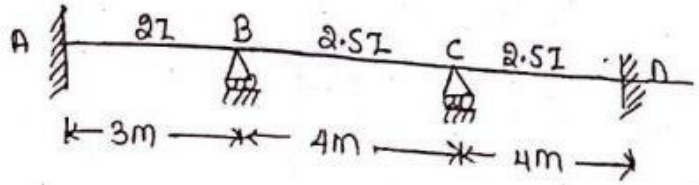

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Fig.-4(b)

(c) Analyse the portal frame ABCD shown in fig.-4(c) using displacement method. 14

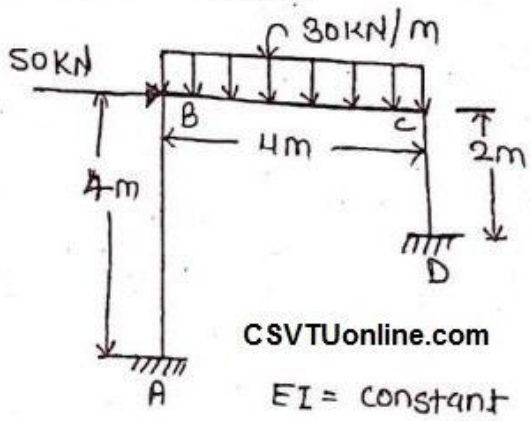

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Fig.-4(c)

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